Qualification Pack

Jr. Designer - Die and Mould for Plastics

QP Code: RSC/Q8002

Version: 3.0

NSQF Level: 5

Rubber, Chemical & Petrochemical Skill development Council || 217, 2nd floor, Rectangle One, Saket District Centre
New Delhi - 110017 || email:vishaka.v@rcpsdc.in

Contents

RSC/Q8002: Jr. Designer - Die and Mould for Plastics

Brief Job Description

The Jr. Designer Die and Mould for Plastics assists in designing, modeling, and documenting dies and moulds for plastic components. The role involves understanding product and tooling requirements, supporting CAD modeling, preparing technical documentation, and performing quality checks under supervision while collaborating with engineers, toolmakers, and production teams.

Personal Attributes

Detail-oriented, analytical, and problem-solving, with good communication and teamwork skills, responsible, organized, safety-conscious, adaptable, and proactive in learning and improving design and documentation practices.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. RSC/N4318: Understand Product and Tooling requirements
2. RSC/N4319: Assist in CAD Modeling of Die and Moulds
3. RSC/N4320: Support in Die and Mould Design Documentation
4. RSC/N5007: Follow health & safety protocol
5. DGT/VSQ/N0102: Employability Skills (60 Hours)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Tool and Die Moulding
Occupation	Tool and Die Designing
Country	India
NSQF Level	5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2144.0803

Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (or equivalent) OR 12th grade Pass (or Equivalent) with 3 Years of experience relevant OR 10th grade pass (or Equivalent) with 6 Years of experience relevant OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Basic knowledge of computer operation
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025
Version	3.0
Reference code on NQR	QG-05-RI-04045-2025-V2-RSDC
NQR Version	2

RSC/N4318: Understand Product and Tooling requirements

Description

This NOS describes the knowledge and skills required to interpret product specifications, drawings, and tooling requirements. It involves identifying the correct tools, materials, and process parameters essential for CNC milling of plastic components, ensuring accuracy and readiness for production.

Scope

The scope covers the following :

- Analyze and Interpret Product Requirements
- Determine Tooling Needs and Specifications
- Collaborate and Communicate Technical Requirements

Elements and Performance Criteria

Analyze and Interpret Product Requirements

To be competent, the user/individual on the job must be able to:

PC1. identify product specifications including dimensions, geometry, material type, and functional requirements

PC2. list out constraints related to manufacturing, cost, and quality standards

PC3. understand compatibility of materials and design with available moulding processes

PC4. point out potential challenges in tooling or production feasibility

PC5. summarize product requirements into clear documentation for design and production teams

PC6. discuss with stakeholders to confirm understanding of product expectations

Determine Tooling Needs and Specifications

To be competent, the user/individual on the job must be able to:

PC7. recognize suitable die and mould types for the product based on geometry and material

PC8. apply knowledge of moulding processes to select cavity configuration, gating system, and cooling channels

PC9. develop design briefs and technical documentation for tooling requirements

PC10. ensure tooling design meets quality, safety, and production standards

PC11. conduct basic checks or simulations to verify tooling feasibility

PC12. state required materials, tolerances, and dimensions for die and mould fabrication

Collaborate and Communicate Technical Requirements

To be competent, the user/individual on the job must be able to:

PC13. discuss tooling requirements with engineers, toolmakers, and production staff

PC14. display and show CAD drawings, prototypes, and specifications effectively

PC15. conduct reviews to ensure tooling meets functional and safety requirements

PC16. mention any adjustments or improvements needed based on stakeholder feedback

PC17. apply documentation standards to communicate technical details clearly

PC18. recognize and report discrepancies or potential design issues during review

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. basics of engineering drawings, symbols, and tolerances
KU2. product specifications such as dimensions, geometry, and material properties
KU3. functional requirements of plastic components in CNC milling and moulding processes
KU4. manufacturing constraints related to cost, cycle time, and machine capability
KU5. different types of moulds, dies, inserts, and their applications
KU6. principles of cavity design, gating systems, cooling channels, and ejection systems
KU7. tooling materials, heat treatment, and surface finishing requirements
KU8. compatibility of plastic materials with different tooling designs
KU9. common tooling defects and preventive measures
KU10. documentation formats for design briefs, technical drawings, and process sheets
KU11. basics of CAD/CAM software used for tooling design and visualization
KU12. industry quality standards (IS/ISO) relevant to tooling and moulds
KU13. workplace safety considerations during tooling design and handling
KU14. environmental and cost considerations in tool selection
KU15. importance of effective communication and stakeholder collaboration in product tooling alignment

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively with engineers, toolmakers, and production staff
GS2. interpret technical drawings, specifications, and CAD models
GS3. document product and tooling requirements in standard formats
GS4. work in teams to review tooling and product compatibility
GS5. use digital tools (e.g., CAD software, spreadsheets) to present technical detail
GS6. follow workplace protocols for reporting issues and documenting reviews
GS7. identify product dimensions, geometry, and materials from work instructions
GS8. analyze constraints related to manufacturing, quality, and cost
GS9. select suitable die/mould type based on product geometry and process requirements
GS10. apply knowledge of moulding processes to recommend cavity configuration and gating system
GS11. define tolerances, materials, and design parameters for tooling
GS12. conduct basic feasibility checks or simulations for tooling designs
GS13. present CAD drawings, prototypes, or samples to stakeholders for review
GS14. suggest adjustments or improvements in tooling design based on feedback

GS15. ensure tooling designs meet safety, quality, and productivity standards

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Analyze and Interpret Product Requirements	**11**	**23**	**-**	**-**
PC1. identify product specifications including dimensions, geometry, material type, and functional requirements	2	4	-	-
PC2. list out constraints related to manufacturing, cost, and quality standards	2	4	-	-
PC3. understand compatibility of materials and design with available moulding processes	2	3	-	-
PC4. point out potential challenges in tooling or production feasibility	2	4	-	-
PC5. summarize product requirements into clear documentation for design and production teams	2	4	-	-
PC6. discuss with stakeholders to confirm understanding of product expectations	1	4	-	-
Determine Tooling Needs and Specifications	**9**	**23**	**-**	**-**
PC7. recognize suitable die and mould types for the product based on geometry and material	1	4	-	-
PC8. apply knowledge of moulding processes to select cavity configuration, gating system, and cooling channels	2	4	-	-
PC9. develop design briefs and technical documentation for tooling requirements	2	4	-	-
PC10. ensure tooling design meets quality, safety, and production standards	1	3	-	-
PC11. conduct basic checks or simulations to verify tooling feasibility	1	4	-	-
PC12. state required materials, tolerances, and dimensions for die and mould fabrication	2	4	-	-
Collaborate and Communicate Technical Requirements	**10**	**24**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss tooling requirements with engineers, toolmakers, and production staff	2	4	-	-
PC14. display and show CAD drawings, prototypes, and specifications effectively	2	4	-	-
PC15. conduct reviews to ensure tooling meets functional and safety requirements	1	4	-	-
PC16. mention any adjustments or improvements needed based on stakeholder feedback	2	4	-	-
PC17. apply documentation standards to communicate technical details clearly	1	4	-	-
PC18. recognize and report discrepancies or potential design issues during review	2	4	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N4318
NOS Name	Understand Product and Tooling requirements
Sector	Rubber
Sub-Sector	
Occupation	Tool and Die Designing
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

RSC/N4319: Assist in CAD Modeling of Die and Moulds

Description

This NOS covers the skills and knowledge required to assist in creating, modifying, and reviewing CAD models of dies and moulds. It includes preparing design inputs, supporting 2D/3D modeling activities, making basic modifications, and ensuring that CAD models align with product specifications, tooling requirements, and quality standards.

Scope

The scope covers the following :

- Understand Product and Tooling Requirements
- Assist in CAD Modeling of Die and Moulds
- Documentation and Technical Communication
- Perform Quality and Design Verification

Elements and Performance Criteria

Understand Product and Tooling Requirements

To be competent, the user/individual on the job must be able to:

PC1. identify functional, aesthetic, and dimensional requirements of plastic products

PC2. list out material properties, product geometry, and manufacturing constraints

PC3. understand the feasibility of different die and mould types for the product

PC4. point out potential production challenges including tooling limitations and cost factors

PC5. Summarize product specifications for documentation and communication with production teams

PC6. discuss requirements with engineers, toolmakers, and supervisors to confirm understanding

Assist in CAD Modeling of Die and Moulds

To be competent, the user/individual on the job must be able to:

PC7. apply CAD software (AutoCAD, SolidWorks, Creo, Fusion 360) to create 2D and 3D models of dies and moulds

PC8. perform modifications on CAD models based on design requirements or feedback

PC9. develop basic assemblies of die and mould components

PC10. show and display CAD models to supervisors and production teams for validation

PC11. carry out measurements and incorporate tolerances accurately in the CAD design

PC12. ensure CAD models comply with product specifications and tooling standard

Documentation and Technical Communication

To be competent, the user/individual on the job must be able to:

PC13. recognize and record tooling specifications, dimensions, and design revisions

PC14. discuss and conduct reviews with engineers, toolmakers, and production staff

PC15. ensure documentation is complete, clear, and accessible for production purposes

PC16. state changes or recommendations required for mould or die improvement

PC17. mention critical features, material considerations, and safety requirements in documentation
PC18. display technical documentation effectively to support production and quality teams

Perform Quality and Design Verification

To be competent, the user/individual on the job must be able to:

PC19. point out potential errors or inconsistencies in CAD models
PC20. apply basic simulation or analysis tools to check feasibility of die and mould design
PC21. perform verification against product specifications and tooling constraints
PC22. develop corrective actions for design discrepancies
PC23. conduct final checks before handing over CAD models for production
PC24. ensure adherence to safety, quality, and dimensional accuracy standards

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. functional, dimensional, and aesthetic requirements of plastic components
KU2. material properties of plastics and their implications for tooling design
KU3. types of dies, molds, and inserts, and their applications in CNC milling/moulding Types of dies, molds, and inserts, and their applications in CNC milling/moulding
KU4. manufacturing constraints related to cost, cycle time, tooling wear, and quality standards
KU5. basics of CAD software (AutoCAD, SolidWorks, Creo, Fusion 360) and their applications
KU6. principles of 2D drafting, 3D modeling, assemblies, and tolerancing in CAD
KU7. design standards, safety requirements, and industry specifications (IS/ISO)
KU8. common tooling and moulding defects and preventive design practices
KU9. documentation standards for recording tooling specifications, dimensions, and revisions
KU10. simulation and analysis tools for checking feasibility of CAD models
KU11. methods of verifying CAD designs against product requirements
KU12. communication protocols for effective interaction with engineers, toolmakers, and supervisors
KU13. workplace safety, data handling, and intellectual property protection in CAD design
KU14. environmental and cost factors in material and tooling selection
KU15. importance of collaboration in cross-functional teams during design reviews

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively with engineers, supervisors, and toolmakers to clarify requirements
GS2. read and interpret engineering drawings, CAD files, and technical documents
GS3. document tooling requirements, revisions, and critical features in standard formats
GS4. conduct discussions and participate in design reviews with stakeholders
GS5. use digital communication tools to share and present CAD models/documents

GS6. maintain accurate records of design changes and approvals
GS7. identify product requirements including geometry, tolerances, and functional needs
GS8. select appropriate die/mould types considering feasibility and material properties
GS9. apply CAD software to create, modify, and assemble die/mould components
GS10. incorporate tolerances, surface finishes, and dimensional accuracy in CAD models
GS11. perform basic simulations or feasibility checks to verify tooling design
GS12. detect and correct errors or inconsistencies in CAD models
GS13. prepare technical documentation including design briefs, tooling specifications, and revision logs
GS14. present CAD models and documentation effectively to production and quality teams
GS15. ensure designs comply with safety, quality, and industry standards before release

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understand Product and Tooling Requirements	**10**	**15**	**-**	**-**
PC1. identify functional, aesthetic, and dimensional requirements of plastic products	2	2	-	-
PC2. list out material properties, product geometry, and manufacturing constraints	2	3	-	-
PC3. understand the feasibility of different die and mould types for the product	1	2	-	-
PC4. point out potential production challenges including tooling limitations and cost factors	2	3	-	-
PC5. Summarize product specifications for documentation and communication with production teams	1	2	-	-
PC6. discuss requirements with engineers, toolmakers, and supervisors to confirm understanding	2	3	-	-
Assist in CAD Modeling of Die and Moulds	**10**	**13**	**-**	**-**
PC7. apply CAD software (AutoCAD, SolidWorks, Creo, Fusion 360) to create 2D and 3D models of dies and moulds	2	1	-	-
PC8. perform modifications on CAD models based on design requirements or feedback	1	2	-	-
PC9. develop basic assemblies of die and mould components	1	2	-	-
PC10. show and display CAD models to supervisors and production teams for validation	2	3	-	-
PC11. carry out measurements and incorporate tolerances accurately in the CAD design	2	3	-	-
PC12. ensure CAD models comply with product specifications and tooling standard	2	2	-	-
Documentation and Technical Communication	**9**	**19**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. recognize and record tooling specifications, dimensions, and design revisions	2	3	-	-
PC14. discuss and conduct reviews with engineers, toolmakers, and production staff	1	2	-	-
PC15. ensure documentation is complete, clear, and accessible for production purposes	1	3	-	-
PC16. state changes or recommendations required for mould or die improvement	1	2	-	-
PC17. mention critical features, material considerations, and safety requirements in documentation	2	3	-	-
PC18. display technical documentation effectively to support production and quality teams	2	6	-	-
Perform Quality and Design Verification	**11**	**13**	**-**	**-**
PC19. point out potential errors or inconsistencies in CAD models	2	2	-	-
PC20. apply basic simulation or analysis tools to check feasibility of die and mould design	2	2	-	-
PC21. perform verification against product specifications and tooling constraints	1	3	-	-
PC22. develop corrective actions for design discrepancies	2	2	-	-
PC23. conduct final checks before handing over CAD models for production	2	2	-	-
PC24. ensure adherence to safety, quality, and dimensional accuracy standards	2	2	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N4319
NOS Name	Assist in CAD Modeling of Die and Moulds
Sector	Rubber
Sub-Sector	
Occupation	Tool and Die Designing
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

RSC/N4320: Support in Die and Mould Design Documentation

Description

This NOS covers the competencies required to assist in preparing, reviewing, and maintaining technical documentation for dies and moulds. It includes recording design specifications, CAD drawings, revisions, and process notes to ensure accuracy, traceability, and compliance with quality, safety, and production standards.

Scope

The scope covers the following :

- Compile and Organize Design Documentation
- Assist in the Preparation and Verification of Technical Documents
- Communication and Collaboration

Elements and Performance Criteria

Compile and Organize Design Documentation

To be competent, the user/individual on the job must be able to:

PC1. identify key design parameters, product specifications, and tooling requirements

PC2. list out material properties, dimensions, tolerances, and manufacturing constraints

PC3. understand standard documentation formats, drawing conventions, and industry requirements

PC4. summarize design changes, revisions, and updates in a clear, organized manner

PC5. point out inconsistencies, missing information, or errors in existing design documents

PC6. discuss the compiled documentation with supervisors or engineers to confirm completeness

Assist in the Preparation and Verification of Technical Documents

To be competent, the user/individual on the job must be able to:

PC7. apply CAD and design software to extract specifications, drawings, and component details for documentation

PC8. develop detailed records of die and mould components, assembly instructions, and operational notes

PC9. ensure accuracy, completeness, and clarity of documentation for production and quality teams

PC10. perform cross-checks to verify design data against product and tooling requirements

PC11. conduct reviews with engineers or supervisors to validate correctness and feasibility of documentation

PC12. state updates or revisions required based on feedback from team members

Communication and Collaboration

To be competent, the user/individual on the job must be able to:

PC13. discuss design details, changes, and updates with engineers, toolmakers, and production staff

PC14. display and show technical drawings, specifications, and design documents for team review
PC15. recognize and mention critical features, safety considerations, and design constraints
PC16. carry out updates and modifications in documentation based on feedback
PC17. ensure proper version control and accessibility of all design documents
PC18. point out potential improvements or discrepancies in design documentation for continuous enhancement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. key product specifications, dimensions, tolerances, and functional requirements of dies and moulds
KU2. material properties and their impact on die/mould design and manufacturing
KU3. standard documentation formats, drawing conventions, and industry best practices
KU4. importance of accuracy, completeness, and clarity in technical documentation
KU5. methods for compiling design revisions, updates, and version control
KU6. principles of CAD/CAM software for extracting design data and drawings
KU7. assembly instructions, operational notes, and component details for dies and moulds
KU8. tools and techniques for cross-checking documentation against product and tooling requirements
KU9. workplace safety, quality standards, and compliance requirements
KU10. communication protocols for discussing documentation with engineers, supervisors, and production teams
KU11. methods for identifying inconsistencies, errors, or missing information in design documents
KU12. importance of traceability and maintaining records for production and quality audits
KU13. collaboration techniques for working with cross-functional teams
KU14. environmental, cost, and operational considerations in design documentation
KU15. continuous improvement practices for documentation management and process enhancement

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively with engineers, supervisors, toolmakers, and production staff
GS2. read and interpret technical drawings, specifications, and CAD documents
GS3. document design revisions, updates, and specifications clearly and accurately
GS4. maintain version control and accessibility of technical documents
GS5. work collaboratively in teams for review and validation of design documentation
GS6. follow workplace protocols for reporting discrepancies or errors
GS7. identify and record key design parameters, product specifications, and tooling requirements
GS8. extract specifications, drawings, and component details using CAD/design software

GS9. compile complete and accurate records of die/mould components, assemblies, and operational notes

GS10. verify documentation against product and tooling requirements for correctness

GS11. update and modify design documents based on feedback from engineers or supervisors

GS12. highlight critical features, safety considerations, and constraints in documentation

GS13. present technical documentation effectively to production and quality teams

GS14. identify potential improvements or discrepancies in design documentation

GS15. ensure compliance with quality, safety, and industry standards in documentation

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Compile and Organize Design Documentation	**13**	**21**	**-**	**-**
PC1. identify key design parameters, product specifications, and tooling requirements	3	3	-	-
PC2. list out material properties, dimensions, tolerances, and manufacturing constraints	2	3	-	-
PC3. understand standard documentation formats, drawing conventions, and industry requirements	2	3	-	-
PC4. summarize design changes, revisions, and updates in a clear, organized manner	2	5	-	-
PC5. point out inconsistencies, missing information, or errors in existing design documents	2	4	-	-
PC6. discuss the compiled documentation with supervisors or engineers to confirm completeness	2	3	-	-
Assist in the Preparation and Verification of Technical Documents	**12**	**19**	**-**	**-**
PC7. apply CAD and design software to extract specifications, drawings, and component details for documentation	2	3	-	-
PC8. develop detailed records of die and mould components, assembly instructions, and operational notes	2	4	-	-
PC9. ensure accuracy, completeness, and clarity of documentation for production and quality teams	3	3	-	-
PC10. perform cross-checks to verify design data against product and tooling requirements	3	3	-	-
PC11. conduct reviews with engineers or supervisors to validate correctness and feasibility of documentation	2	3	-	-
PC12. state updates or revisions required based on feedback from team members	2	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Communication and Collaboration	**15**	**20**	**-**	**-**
PC13. discuss design details, changes, and updates with engineers, toolmakers, and production staff	3	3	-	-
PC14. display and show technical drawings, specifications, and design documents for team review	2	3	-	-
PC15. recognize and mention critical features, safety considerations, and design constraints	3	3	-	-
PC16. carry out updates and modifications in documentation based on feedback	2	4	-	-
PC17. ensure proper version control and accessibility of all design documents	2	4	-	-
PC18. point out potential improvements or discrepancies in design documentation for continuous enhancement	3	3	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N4320
NOS Name	Support in Die and Mould Design Documentation
Sector	Rubber
Sub-Sector	
Occupation	Tool and Die Designing
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor

PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy

PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc

PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.

PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use

PC6. dispose off waste safely and correctly in a designated area

PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace

PC8. perform work in a manner which minimizes environmental damage

PC9. all procedures and work instructions for controlling risk are followed closely

PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.

PC12. follow emergency procedures as per company standards and workplace requirements

PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements

PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.

PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate

PC16. dispose off medical waste in accordance with workplace requirements

PC17. report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

PC18. comply with general safety procedures

PC19. follow standard safety procedures while handling equipment, hazardous material or tool

PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.

PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure

PC22. keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

PC23. attend fire drills and other safety related workshops organized at the workplace

PC24. be aware of first aid, evacuation and emergency procedures

PC25. be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)

PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. policies on incentives, delivery standards, and personnel management.

KU2. occupational safety and health policy followed

KU3. emergency evacuation procedure

KU4. medical policy

KU5. company laws and acts

KU6. the risks to health and safety and the measures to be taken to control those risks in the area of work

KU7. workplace procedures and requirements for the handling of workplace injuries/illnesses.
KU8. basic emergency first aid procedure
KU9. local emergency services
KU10. reporting on accidents, incidents and problems to appropriate authorities.
KU11. how to use machines as per standard operating procedure
KU12. how to maintain work area safe and secure
KU13. use of hazardous materials, tools and equipments
KU14. emergency evacuation and first aid procedures to be followed
KU15. personal hygiene and fitness requirements
KU16. general duties under the relevant health and safety legislation
KU17. what personal protective equipment and clothing should be worn and how it is cared for
KU18. the correct and safe way to use materials and equipment required for work
KU19. the importance of good housekeeping in the workplace
KU20. safe disposal methods for waste
KU21. methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. record data which are required for record keeping purpose
GS2. report problems to the appropriate person in a timely manner
GS3. write descriptions and details about incidents in reports
GS4. read instruction manuals for hand tools and equipment
GS5. read instructions on work orders and procedures
GS6. receive instructions and seek advice from superiors
GS7. communicate clearly and effectively with others
GS8. take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
GS9. work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
GS10. make changes in cycle time due to improved process.
GS11. use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
GS12. consult the peer group and superiors to arrive at a favourable decision.
GS13. use of standard available problem solving techniques for decision making
GS14. review and analyze the process steps to check on system non adherence andnon conformity
GS15. review the current sop and other standards for continuous improvement to facilitate decision making
GS16. take a calculated risk with minimum losses
GS17. schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

GS18. match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)

GS19. ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.

GS20. complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.

GS21. communicate effectively to the superior/customer for any delay in supplies to the clients.

GS22. work towards fulfilling the customers requirement as per their demand.

GS23. in case of any complaint, ensure its timely resolution if the problem is emanating at his level

GS24. communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.

GS25. maintain good/cordial relation with customers.

GS26. work on the feedback received from customer regarding the product.

GS27. use first aid treatment in case of any injury/accident.

GS28. monitor and maintain the condition of tools and equipment

GS29. assess situation & identify appropriate control measures

GS30. act, communicate and report in emergency situation

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Maintain a clean and efficient workplace	**9**	**21**	**-**	**-**
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
Render appropriate emergency procedures	**9**	**18**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
Maintain standard safety procedures at the workplace	**6**	**15**	**-**	**-**
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
Participate in safety awareness campaigns	**2**	**8**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
Understand potential sources of accidents	**1**	**3**	**-**	**-**
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
Use safety gears to avoid accidents	**3**	**5**	**-**	**-**
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N4318.Understand Product and Tooling requirements	30	70	-	-	100	20
RSC/N4319.Assist in CAD Modeling of Die and Moulds	40	60	-	-	100	25
RSC/N4320.Support in Die and Mould Design Documentation	40	60	-	-	100	25
RSC/N5007.Follow health & safety protocol	30	70	-	-	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15
Total	**160**	**290**	**-**	**-**	**450**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.